Is the Equal Area Criterion Applicable for Grid-Forming (GFM) Inverters?

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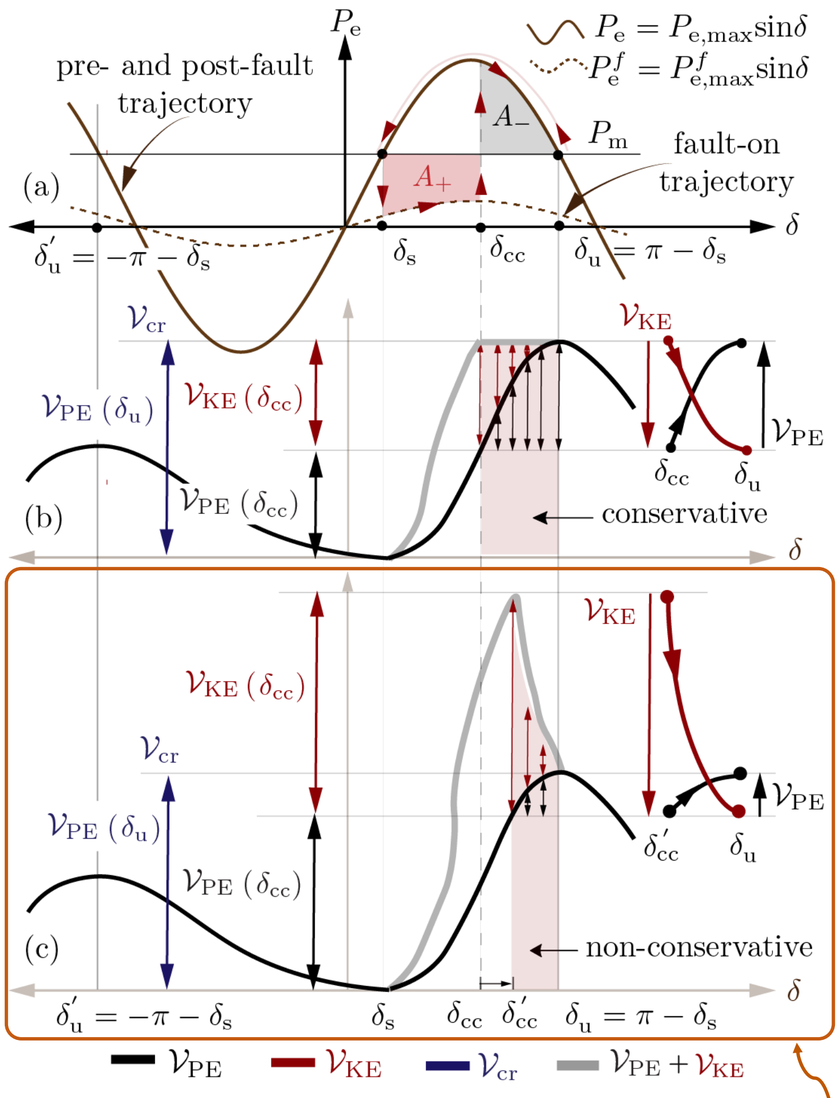
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Motivation

- Classical transient stability assessment (TSA) methods can't capture non-conservative damping effects in GFMs
- Multi-GFM systems face complex interactions (e.g., non-uniform damping) that existing techniques can't address
- Understanding how damping impacts GFM systems is essential to develop tailored solutions

Classical TSA Methods and Damping

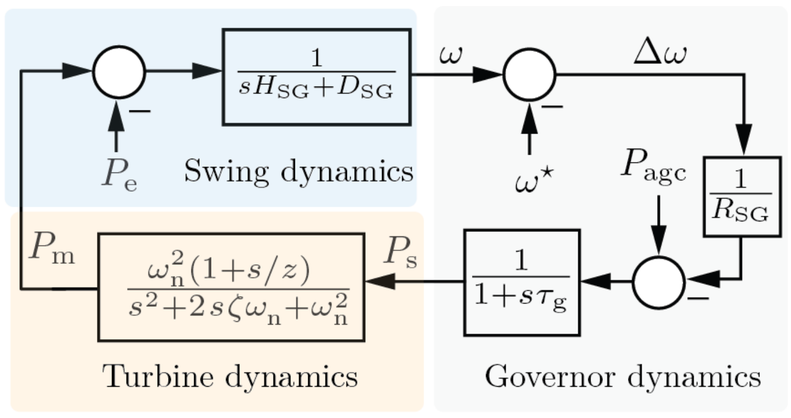


In (c) above, part of the total energy is dissipated with damping so that the critical clearing angle is increased

$$\mathcal{V}_{PE}(\delta_{cc}) + \mathcal{V}_{KE}(\omega_{cc}) - \boxed{\mathcal{V}_d} \leq \mathcal{V}_{cr}$$

Dynamics of Machines and GFM Inverters

The frequency dynamics of machines are governed by the speed governor, turbine, and swing equations



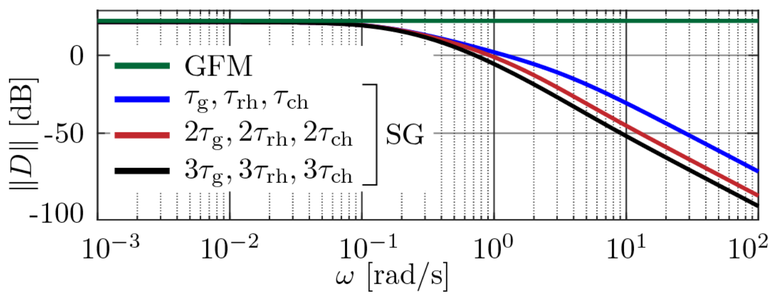
The frequency dynamics of the SGs and GFM inverters can be represented by the common equation:

$$\Delta P = H\ddot{\delta} + D\dot{\delta}$$

	Synchronous generator	GFM droop inverter
ΔP	$P^\star G(s) - P_e$	$P^\star - P_e$
H	H_{SG}	$\frac{1}{\omega_c R_{GFM}}$
$\boxed{D}$	$\cancel{D_{SG}} + \frac{1}{R_{SG}}\boxed{G(s)}$	$\frac{1}{R_{GFM}}$

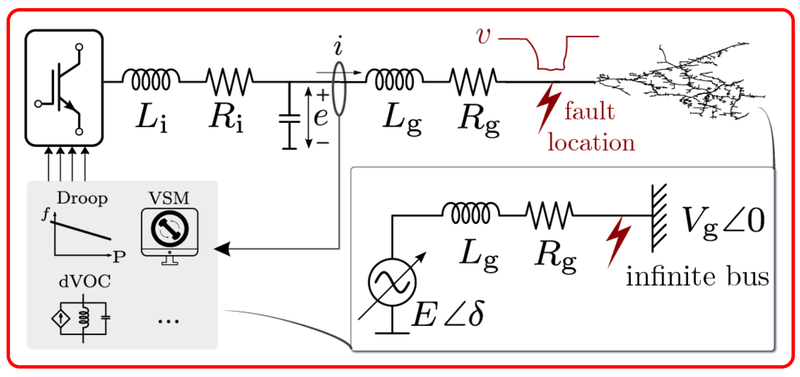
For SG, magnitude of G(s) decreases significantly after 0.1 rad/sec, rendering the damping term ineffective during short-term disturbances

For GFM inverters, damping is the inverse of droop gain and there is no frequency dependency

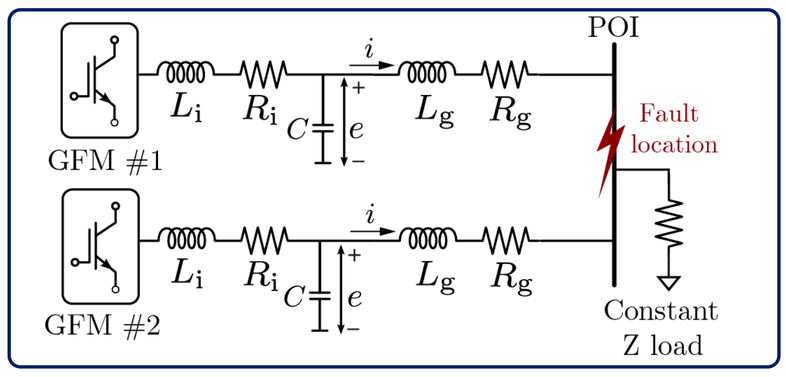


For GFM inverters, constant damping during transient invalidates the conservativeness assumption

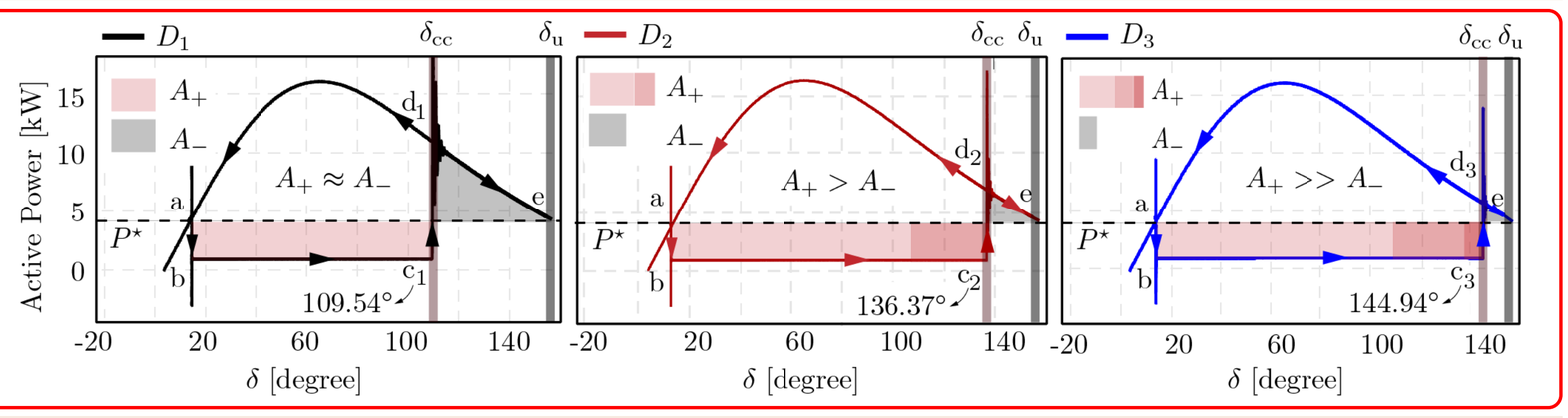
Test System with Single GFM Inverter



Test System with Parallel GFM Inverters



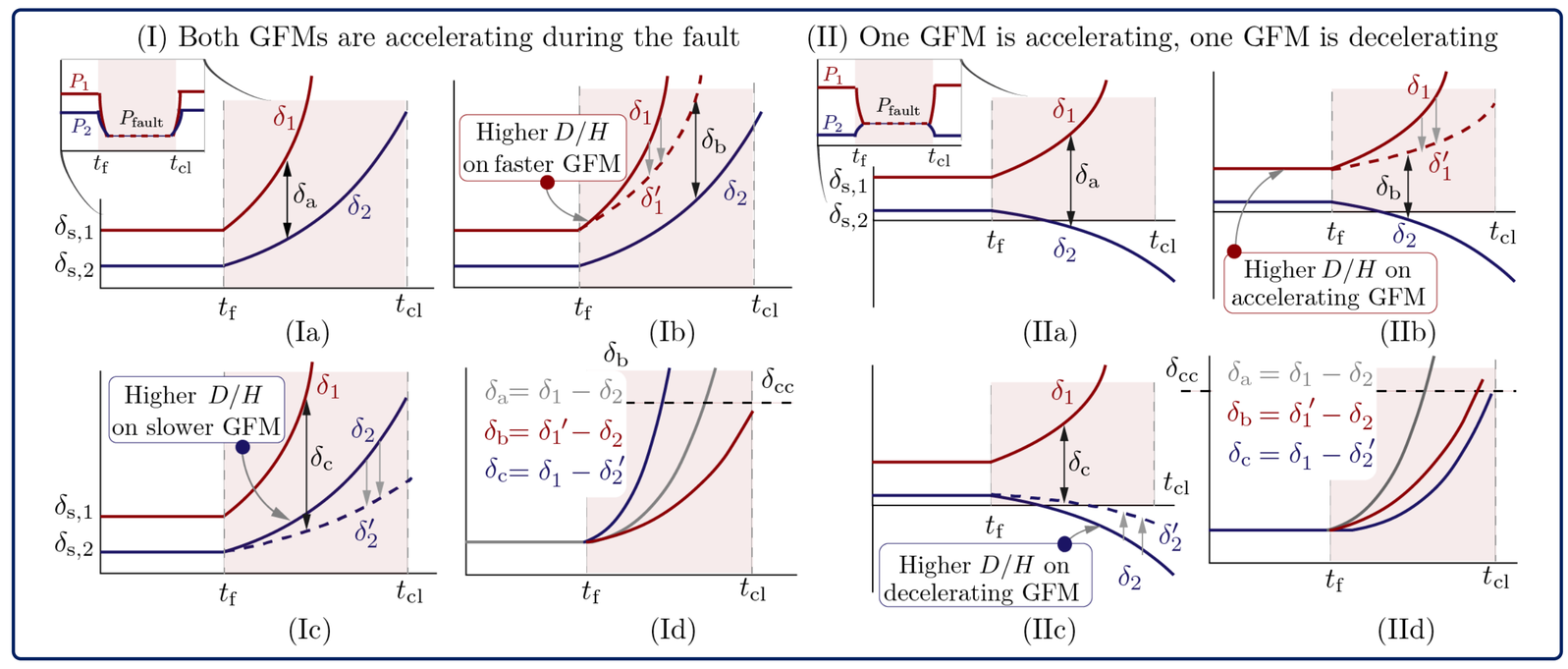
Simulation Results I: Single GFM Inverter (D_1<D_2<D_3)



- Increasing damping causes a noticeable imbalance between the acceleration and deceleration areas
- Notably, in all cases, power angle returns to the stable operating point, clearly violating the EAC

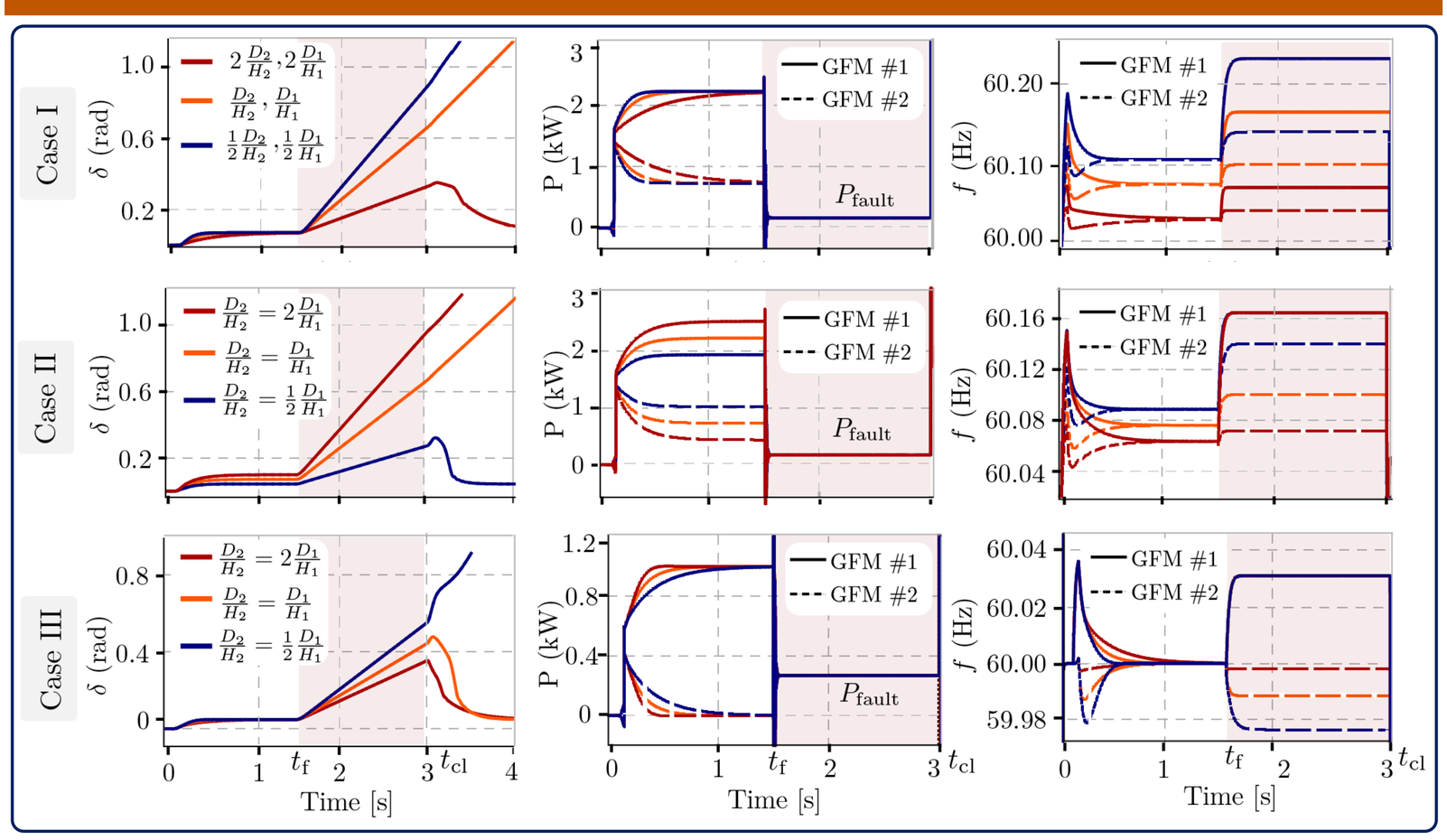
Impact of Damping on Parallel GFM Systems

- Multi-GFM systems may show complex interaction due to heterogeneous parameters
- Unlike single-GFM system, increasing damping does not necessarily enhance stability margin



(I) GFM inverters accelerate during the fault; **(II)** one GFM inverter accelerates while the other decelerates. **(a)** shows the base case. **(b)–(c)** illustrate changes in angle dynamics when a higher damping-to-inertia ratio, D/H, is applied to one of the GFM inverters. **(d)** shows the relative angles for the three cases.

Simulation Results II: Parallel-connected GFM Inverters



Case I compares three scenarios where D/H is increased, decreased, or kept at the base level for both inverters. **Cases II and III** explore the impact of altering D/H for only one inverter (GFM #2): in **Case II,** both inverters accelerate during the fault, while in **Case III,** one inverter accelerates and the other decelerates. In **Case II,** D/H is changed for the slower inverter, while in **Case III,** D/H is altered for the decelerating inverter.

Conclusion and Future Work

- Damping significantly improves stability margins in single-inverter systems, but non-uniform damping may destabilize multi-inverter systems
- Future work will focus on modifying classical TSA approaches to incorporate damping effects

[1] D. Chatterjee, B. Johnson, and Gab-Su Seo, "Is Equal Area Criterion Applicable for Transient Stability Assessment of Grid-Forming Inverters?," submitted to IEEE PESGM 2025

[2] D. Chatterjee, Gab-Su Seo, and B. Johnson, "Stability Analysis of Multi-inverter Systems with Heterogeneous Grid-Forming Dynamics," submitted to IEEE PESGM 2025